THE FLATNESS FUNCTOR IS AN FPQC SHEAF AND LIMIT PRESERVING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Flatness of the base change along a surjective flat morphism, and descent of flatness for a finite presentation module. More on Morphisms, label more-morphisms-lemma-flat-locus-base-change (proof line 6456); Limits, label limits-lemma-descend-module-flat-finite-presentation (proof line 6458). The first shows that flatness of the pulled back module along a surjective flat morphism is equivalent to flatness of the original, giving the sheaf property, and the second gives the limit preserving statement, following the same reduction as the dimension functor lemma.

Source reference. Original lemma 05MV: sources/stacks-flat/pinned-original-source.tex, source lines 6337–6337. Navigation pointer for \ref{lemma-flat-dimension-n} used inside the counted span (line 6461); the argument that consumes it is retained in the span.

Source reference. Original lemma 05LY: sources/stacks-chow/tags.txt, source lines 7252–7252. Navigation pointer for \ref{limits-lemma-descend-module-flat-finite-presentation} used inside the counted span (line 6458); the argument that consumes it is retained in the span.

Source reference. Original lemma 047C: sources/stacks-chow/tags.txt, source lines 5503–5503. Navigation pointer for \ref{more-morphisms-lemma-flat-locus-base-change} used inside the counted span (line 6456); the argument that consumes it is retained in the span.

Source reference. Original reference 05MW: sources/stacks-flat/pinned-original-source.tex, source lines 6421–6421. Navigation pointer for \ref{situation-flat} used

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inside the counted span (line 6443); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 05MW.*

(1) *The functor F_{flat} satisfies the sheaf property for the fpqc topology.*
(2) *If f is quasi-compact and locally of finite presentation and $\mathcal{F}$ is of finite presentation, then the functor F_{flat} is limit preserving.*

Proof. Part (1) follows from the following statement: If $T' \to T$ is a surjective flat morphism of schemes over S, then $\mathcal{F}_{T'}$ is flat over T' if and only if $\mathcal{F}_T$ is flat over T, see More on Morphisms, Lemma 047C. Part (2) follows from Limits, Lemma 05LY after reducing to the case where X and S are affine (compare with the proof of Lemma 05MV). □